

NeuroSleep: A 99,477-Parameter Multi-Modal PSG Model for Five-Stage Sleep Scoring on Resource-Constrained Devices

A detailed model and evaluation paper based on the public NeuroSleep repository and supporting literature

Kaushik P. | Suha Vora | Shailendra Bhatt | Shamique Khan | Aasir Jaffer Lone | Prachi Kamboj

VIT Bhopal University

Model	Parameters	Context	Inputs	Outputs
ImprovedStudent	99,477	10 x 30 s = 300 s	4 PSG channels	W / N1 / N2 / N3 / REM

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Abstract

Background: Automatic sleep staging transforms long polysomnography (PSG) recordings into a sequence of sleep-stage labels. Modern systems can be highly accurate but may be computationally expensive, data-hungry, or difficult to deploy outside a laboratory. **Objective:** This paper documents NeuroSleep, a compact four-channel PSG model designed around a 99,477-parameter budget and a 300-second causal context. **Methods:** The current ImprovedStudent architecture combines a multi-resolution temporal convolutional stem, two depthwise-separable convolutional blocks, adaptive pooling, a parametric eight-filter Gabor feature bank with learnable frequency and bandwidth, a two-layer GRU with 64 hidden units, and a five-class linear head. Inputs are 10 consecutive 30-second epochs from Fpz-Cz EEG, Pz-Oz EEG, EOG, and EMG at 100 Hz. The repository reports 10-fold person-level cross-validation over 92 eligible records from 52 persons and three random seeds, with causal last-epoch evaluation. **Results:** The primary benchmark reports stage-wise F1 of 0.960 for Wake, 0.445 for N1, 0.733 for N2, 0.700 for N3, and 0.782 for REM. A separate standalone exhibition run reports 90.57% accuracy, Cohen kappa 0.8080, macro F1 0.7490, and 6.2 ms/batch CPU latency. **Conclusion:** The model demonstrates a practical design point between multi-modal physiological context and strict compute constraints. Its main research value is the explicit coupling of temporal multi-resolution filtering, parameter-efficient convolution, learnable spectral structure, and recurrent context. Major remaining limitations are small effective person count, class imbalance, lack of external-dataset validation, and artifact-version discrepancies that must be resolved before publication-quality benchmarking.

Keywords: sleep staging; polysomnography; EEG; EOG; EMG; deep learning; GRU; depthwise-separable convolution; Gabor filters; edge AI; resource-constrained inference; Sleep-EDF

Research Scope and Evidence Basis

The paper treats the current repository state as the system of record for the reported architecture and primary benchmark. The repository description identifies NeuroSleep as a five-stage sleep classifier using four PSG channels, a 300-second context window, and a 99,477-parameter ImprovedStudent model. The repository also states that its person-level benchmark is based on 92 eligible records from 52 persons, with three seeds and ten person-level folds. [1]

Two artifact-version issues are documented rather than silently reconciled. First, the README and the hosted Hugging Face model card report different benchmark numbers for the hosted 99K model. Second, the README and configuration text differ on batch size and on some component-level parameter accounting. This paper therefore reports the repository README benchmark as the primary result and records the differences in the reproducibility section rather than mixing figures from separate artifacts. [1,2]

The repository further states that an adaptation study using frozen, LoRA, and full-fine-tuning variants was contaminated by record-level/person-level leakage and should not be treated as valid generalization evidence. Those results are excluded from all performance conclusions in this paper. [1]

1. Introduction

Sleep staging is the process of dividing overnight PSG into standardized epochs and assigning each epoch a stage such as Wake, N1, N2, N3, or REM. Automated staging is attractive because manual scoring is labor-intensive and because long recordings contain temporal structure that can be modeled computationally. AASM tooling exposes the five-stage scoring vocabulary and uses a default 30-second epoch view for stage scoring. [3]

The modeling challenge is non-trivial. EEG contains cortical oscillatory information at multiple temporal scales; EOG supplies eye-movement signatures that help distinguish REM and wake-related transitions; EMG provides muscle-tone information that is particularly useful for REM characterization. At the same time, sleep-stage labels are highly imbalanced, transitions are structured, and the clinically difficult N1 stage is relatively infrequent. Modern reviews emphasize that evaluation quality depends strongly on

dataset choice, preprocessing, architecture, validation protocol, and metric selection, with overall accuracy potentially hiding minority-stage failure. [4]

The design of NeuroSleep addresses these constraints through a deliberately small architecture. Rather than using a large transformer or image-like spectrogram pipeline, the current model keeps the raw multi-channel time-series representation, extracts local and longer-window temporal features with two convolutional branches, injects learnable frequency-selective structure with a compact Gabor bank, and uses a small GRU to model 300 seconds of sequential context. The result is a 99,477-parameter network intended for CPU-class or embedded deployment. [1]

2. Research Question and Design Objectives

The central engineering question is: can a small multi-modal model retain enough intra-epoch and inter-epoch information to perform five-stage sleep staging while remaining practical for resource-constrained deployment?

O1 - Multi-modal utilization. Use EEG, EOG, and EMG jointly rather than relying on a single biological source.

O2 - Multi-scale representation. Capture fast transients and slower oscillatory structure within each 30-second epoch.

O3 - Spectral inductive bias. Expose learnable frequency-selective features without constructing a large spectrogram branch.

O4 - Temporal context. Model stage continuity and transitions using a 10-epoch, 300-second recurrent context.

O5 - Edge efficiency. Constrain parameter count and runtime while keeping the complete scoring pipeline deployable on CPU-oriented hardware.

O6 - Person-level validity. Prevent subjects from appearing in both training and test folds and evaluate the unique final epoch of each causal sequence.

End-to-end methodological pipeline

The training and evaluation protocol is designed to separate person identities across test folds and report the causal last-epoch prediction.

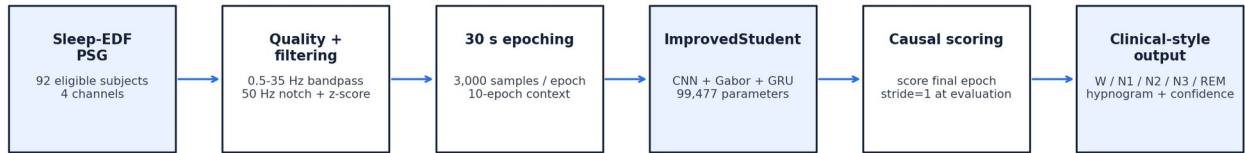


Figure 1. End-to-end NeuroSleep pipeline. The figure reflects the current repository protocol and distinguishes the model from historical/adaptation experiments.

3. Dataset and Sleep-Staging Task

The system uses the Sleep-EDF Expanded database hosted by PhysioNet. The full public database contains 197 whole-night PSG recordings with EEG, EOG, chin EMG, and event-marker channels. The model pipeline, however, reports an eligible cohort of 92 records after excluding eight downloaded subjects that were wake-only in the project processing pipeline. Those records correspond to 52 unique persons, and both nights are grouped at the person level during cross-validation. The dataset provides Fpz-Cz and Pz-Oz EEG channels and EOG/EMG signals sampled at 100 Hz, consistent with the model input contract. [1,5]

Item	Reported value	Interpretation
Source	Sleep-EDF Expanded (PhysioNet)	Whole-night PSG with expert annotations
Downloaded subjects	100	Project acquisition pool reported by repository
Excluded	8 wake-only subjects	No usable sleep-stage sequence for the target task
Eligible records	92	Evaluation cohort
Unique persons	52	Person identity is the unit of test separation
Channels	Fpz-Cz, Pz-Oz, EOG, EMG	2 EEG + ocular + muscle activity
Sampling rate	100 Hz	3,000 samples per 30-second epoch
Stage labels	Wake, N1, N2, N3, REM	Five-class AASM-style target space
Context	10 epochs / 300 seconds	Five-minute temporal receptive context

The class distribution reported by the repository is approximately 68% Wake, 4.6% N1, 17% N2, 6% N3, and 6% REM. The skew makes a single accuracy number insufficient for scientific interpretation; the paper therefore emphasizes stage-wise F1 and Cohen kappa alongside macro/weighted F1 where available. [1,4]

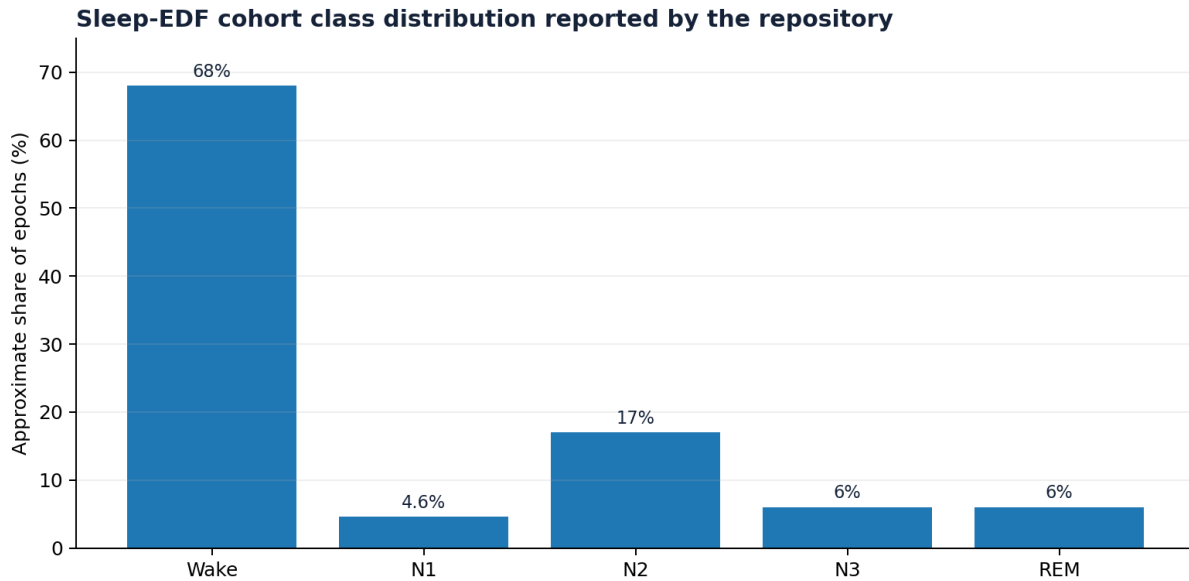


Figure 2. Approximate class distribution reported by the repository. The severe N1 minority is an important source of modeling difficulty.

4. Signal Representation and Preprocessing

The hosted model card associated with the current 99K checkpoint documents the following preprocessing chain: 0.5-35 Hz band-pass filtering, 50 Hz notch filtering, per-channel z-score normalization, and 30-second epoching at 100 Hz. The source repository separately exposes dedicated filtering and signal-quality modules. Because the repository README does not restate every filter setting, the exact frequencies are attributed to the hosted model card rather than presented as an independently reconstructed source-code fact. [1,2]

Stage	Operation	Purpose
1	0.5-35 Hz band-pass	Retain major EEG/EOG/EMG content used for stage discrimination while attenuating slow drift and high-frequency noise
2	50 Hz notch	Suppress mains interference in regions where line noise is expected
3	Per-channel z-score	Standardize amplitude scale and reduce subject/channel gain variation
4	30-second epoching	Align inputs with standard sleep-scoring epochs
5	10-epoch windowing	Provide 300 s temporal context to the recurrent classifier
6	Quality control	Repository exposes signal-quality utilities; exact thresholds are implementation-specific and should be read from the pinned code/config for reproduction

Tensor contract: a training/inference sequence is represented as X in $\mathbb{R}^{(B \times 10 \times 4 \times 3000)}$. The model returns Y in $\mathbb{R}^{(B \times 10 \times 5)}$. The deployment example applies softmax over the five logits and selects the argmax stage for each epoch. [1,2]

5. Proposed ImprovedStudent Architecture

The current production-oriented model contains five conceptual stages: (i) multi-resolution temporal stem, (ii) depthwise-separable CNN encoder, (iii) adaptive pooling plus learnable Gabor feature extraction, (iv) two-layer GRU sequence modeling, and (v) a linear five-class head. The entire network contains 99,477 trainable parameters according to the repository. [1]

NeuroSleep: current ImprovedStudent architecture

A multi-modal, resource-constrained ANN for causal five-stage sleep scoring

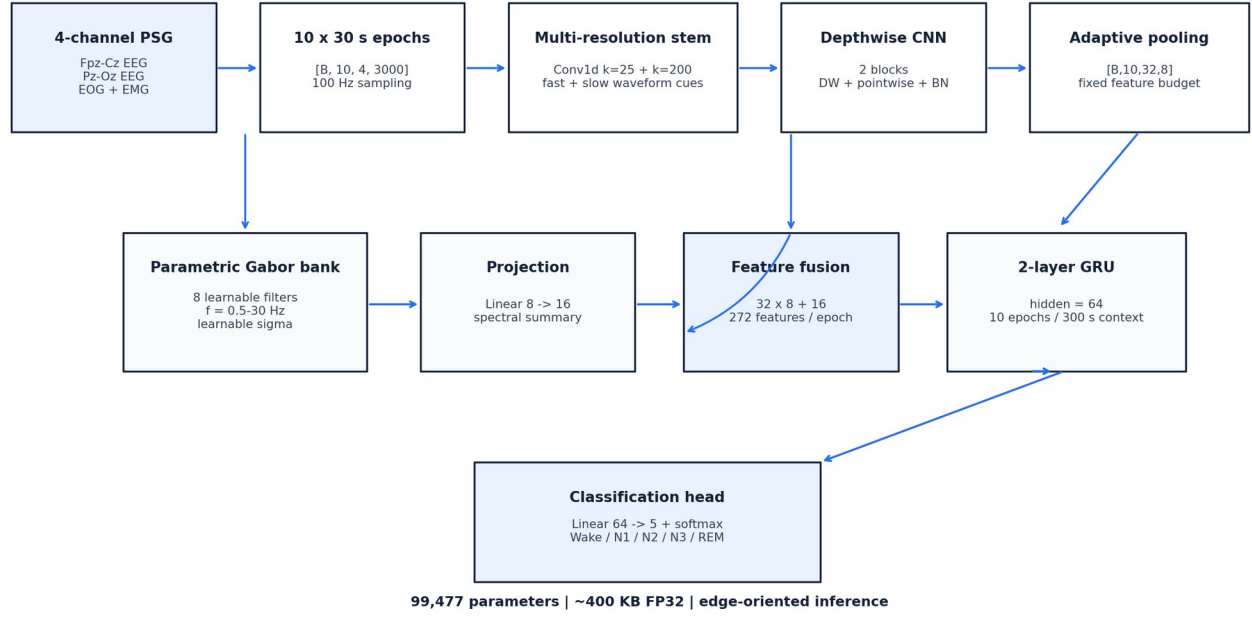


Figure 3. Detailed architecture of the current ImprovedStudent model. Dimensions shown are those documented by the repository; the stem output is approximately 500 samples along the temporal axis.

5.1 Multi-resolution convolutional stem

Two parallel one-dimensional convolutional branches process each epoch: a short kernel of 25 samples and a long kernel of 200 samples. At 100 Hz, these correspond to approximately 250 ms and 2.0 s temporal windows. The short branch can respond to fast waveform changes, while the long branch provides a broader temporal view of oscillatory structure. Their outputs are concatenated into a compact feature representation of approximately 16 channels and about 500 temporal positions. [1]

This design encodes a useful inductive bias for sleep signals: the model does not have to discover every temporal scale from a single convolutional receptive field. Spindles, K-complex-like transients, eye-movement bursts, and slow-wave activity occupy different temporal scales; parallel filters make those scales explicit without requiring a large feature pyramid.

5.2 Depthwise-separable convolutional encoder

The encoder contains two depthwise-separable convolutional blocks. Each block applies a depthwise temporal convolution followed by a pointwise 1×1 convolution and batch normalization. Depthwise convolution applies spatial/temporal filtering independently to each feature map, while pointwise convolution mixes information across feature channels. This factorization reduces multiply-add cost relative to a dense convolution with the same channel width and is a standard strategy for compact neural networks.

The model uses $16 \rightarrow 32$ channels in the first block and $32 \rightarrow 32$ in the second. The design concentrates most parameters elsewhere: the repository reports only 1,904 parameters for the encoder, compared with 89,856 in the GRU. This means the encoder is intentionally compute-light and the recurrent layer is the dominant capacity allocation. [1]

5.3 Adaptive average pooling

After convolutional encoding, adaptive average pooling maps the temporal feature sequence to a fixed 8-position representation for every epoch, producing a tensor described as $[B, 10, 32, 8]$. Adaptive pooling is useful here because the downstream recurrent layer should see a fixed feature dimensionality regardless of small implementation-level changes in convolutional temporal extent. It also limits the activation footprint before feature fusion. [1]

5.4 Parametric Gabor feature extraction

The model adds an explicit learnable spectral branch rather than relying only on unconstrained convolutional filters. Eight learnable Gabor-like filters are parameterized by center frequency and bandwidth (sigma), with frequencies constrained or initialized within approximately 0.5-30 Hz. The resulting spectral descriptors are projected with a small $\text{Linear}(8 \rightarrow 16)$ layer and fused with the pooled CNN features. [1,2]

For intuition, a simplified zero-phase Gabor kernel can be written as $g_k(t) = \exp(-t^2 / (2 \sigma_k^2)) \cos(2 \pi f_k t)$. Here f_k controls center frequency and σ_k controls effective bandwidth. The implementation is not presented as a fixed handcrafted filter bank: the important feature is that the spectral parameters are learned jointly with the classifier.

The resulting per-epoch feature width is reported as $272 = (32 \times 8) + 16$. The first term preserves compact CNN feature structure, while the second adds an explicitly frequency-oriented summary. This hybrid representation is especially appropriate for sleep staging because canonical sleep physiology is strongly frequency-dependent, while stage decisions also depend on morphology and cross-channel events.

5.5 Two-layer GRU temporal model

The 272-dimensional per-epoch vectors are passed through a two-layer GRU with 64 hidden units. The GRU maintains a hidden state across the 10 epochs, giving the classifier access to 300 seconds of temporal context. The model can therefore learn stage persistence and transition structure instead of classifying each 30-second epoch as an independent sample. [1]

For a sequence $z_1, \dots, z_T$, the recurrent state can be abstracted as $h_t = \text{GRU}(z_t, h_{t-1})$, followed by logits $y_t = W h_t + b$. The final linear layer maps 64 hidden features to five logits. Softmax converts logits to class probabilities. In the causal evaluation protocol, only the final epoch in each 10-epoch window contributes to the evaluation target, which avoids counting the same unique epoch repeatedly across overlapping contexts.

5.6 Classification head

The output layer is Linear(64 $\rightarrow$ 5). The five logits correspond to Wake, N1, N2, N3, and REM. The head contains only 325 parameters, making it negligible in the total model budget. The compact head also makes the class semantics explicit and keeps inference deterministic once the feature sequence has been computed.

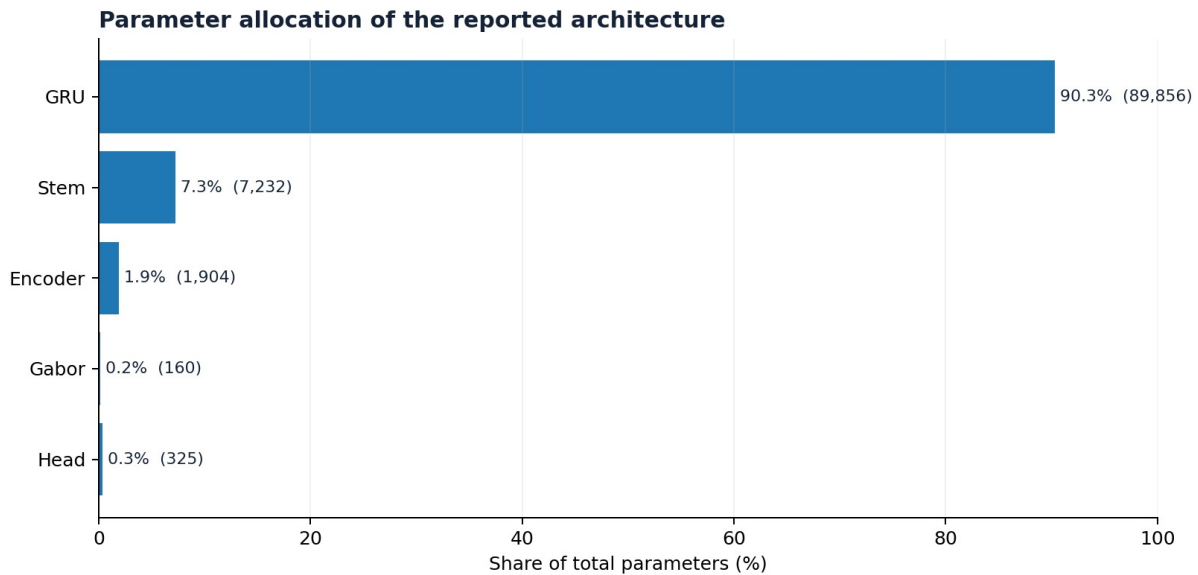


Figure 4. Parameter allocation reported in the repository. The GRU accounts for approximately 90.3% of the stated 99,477-parameter budget.

6. Why These Components Are Used

Component	What it learns	Why it matters for sleep staging	Efficiency implication
Fpz-Cz / Pz-Oz EEG	Cortical rhythms and morphology	Separates wake, NREM depth, and REM-related signatures	Raw waveform input avoids large image-like spectrogram tensors
EOG	Eye movements and ocular activity	Helpful for REM and wake transitions	One extra channel adds little parameter cost
EMG	Muscle tone	Supports REM discrimination and wake/muscle activation cues	Compact channel-level fusion
25-sample Conv1d	Short temporal patterns	Fast transients and brief waveform events	Small receptive field
200-sample Conv1d	Longer temporal patterns	Slow oscillatory structure	Adds scale without a deep pyramid
Depthwise-separable CNN	Local temporal features with channel mixing	Efficient nonlinear representation	Lower parameter/compute cost
Gabor bank	Learnable frequency-selective features	Makes spectral structure explicit	Only eight filters plus small projection
2-layer GRU	Inter-epoch dependencies	Models continuity and transitions over 5 minutes	Capacity concentrated in one compact recurrent module
Linear 64 $\rightarrow$ 5	Stage decision	Maps sequence features to AASM-style class space	Negligible parameter footprint

7. Training Objective and Optimization

The repository reports supervised training with categorical cross-entropy and class weighting. For a target stage y_t and predicted probability $p_{t,y}$, a weighted sequence loss can be written as $L = -(1/T) \sum_t w_{(y_t)} \log p_{(t,y_t)}$. The documented weights double the contribution of N1 and REM, addressing their lower prevalence in the reported Sleep-EDF cohort. [1,2]

Hyperparameter	Repository-reported setting
Optimizer	AdamW
Learning rate	3×10^{-4}
Weight decay	1×10^{-4}
Training duration	20 epochs
Early stopping	Patience 5
Scheduler	Cosine annealing; config additionally states 10% warmup
Gradient clipping	max_norm = 1.0
Mixed precision	CUDA AMP
Class weights	N1 = 2x, REM = 2x
Batch size	README benchmark summary: 32; model config: 16 (artifact discrepancy)
Evaluation seeds	42, 43, 44

AdamW is appropriate for this setting because decoupled weight decay can regularize compact networks without coupling the decay term to the adaptive moment estimates. More importantly, the project uses a deterministic protocol fingerprint that records the dataset manifest hash, checkpoint hash, config hash, Git commit, software versions, GPU identity, and training hyperparameters. That design is unusually useful for a student-scale research project because it turns reproducibility from a narrative claim into a testable artifact. [1]

8. Evaluation Protocol

The principal evaluation is a person-level 10-fold cross-validation procedure over 52 unique persons, repeated with seeds 42, 43, and 44 for a total of 30 fold-seed evaluations per method. Whole persons, including both nights where present, are kept within a single test fold. This is materially stronger than record-level splitting because subject-specific waveform characteristics cannot leak from training into test examples. [1]

Training supervision. The repository reports all-position supervision during training using stride=5 windows. This means multiple epoch positions within a training window provide learning signal, increasing training efficiency without changing the causal test contract.

Primary evaluation. Testing uses stride=1 and supervises only the last epoch of a 10-epoch context. The design is causal: the model receives the current target epoch plus preceding context, and no future epochs are available.

Metrics. Accuracy measures overall correctness; Cohen kappa measures chance-corrected agreement; macro F1 weights each stage equally; weighted F1 weights stages by support; the repository also exposes geometric mean of per-class recalls (MGm).

9. Primary Benchmark Results

The current repository reports the following person-level stage-wise F1 results across the 30 fold-seed evaluations. These are the primary scientifically relevant results in this paper. [1]

Stage	F1 summary	95% CI	Interpretation
Wake	0.960 ± 0.026	[0.950, 0.970]	Strong separation of awake epochs
N1	0.445 ± 0.087	[0.413, 0.478]	Most difficult class; minority and transitional
N2	0.733 ± 0.163	[0.672, 0.794]	Moderate-to-strong performance on prevalent N2
N3	0.700 ± 0.112	[0.658, 0.742]	Captures deep-sleep patterns despite lower support
REM	0.782 ± 0.116	[0.739, 0.826]	Good performance on a physiologically distinctive stage

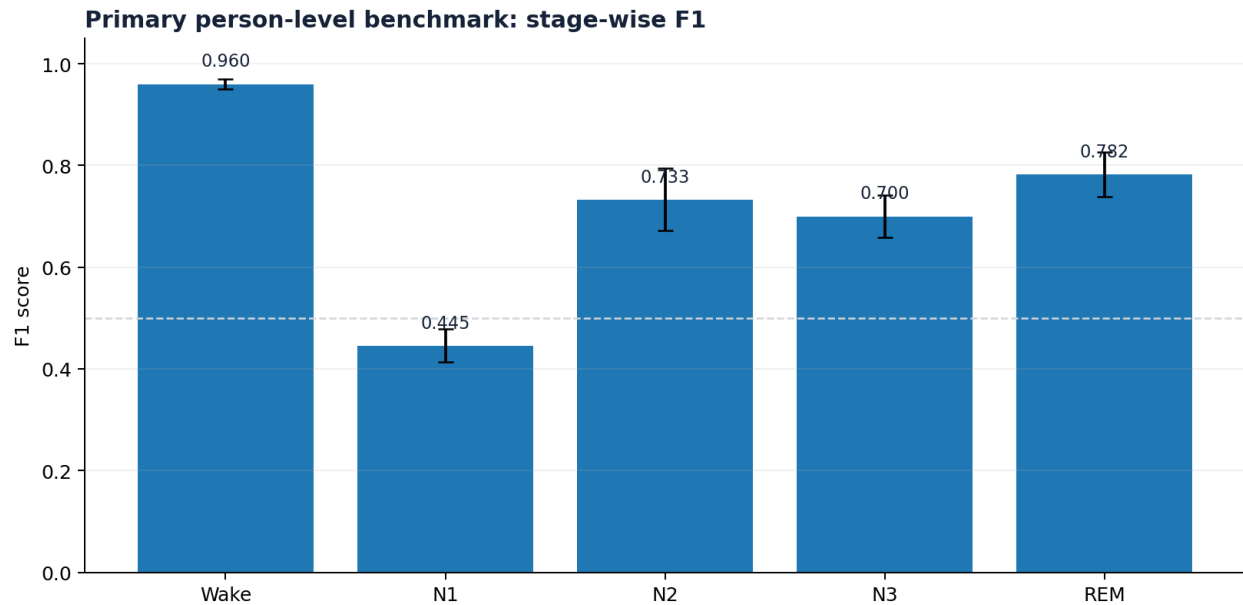


Figure 5. Per-stage F1 with the repository-reported 95% confidence intervals for the primary person-level benchmark.

The most consequential pattern is not the high Wake score alone but the gap between common and transitional stages. N1 remains substantially harder than N2/N3/REM, consistent with its low prevalence and physiological overlap with Wake and N2. The repository itself attributes N1 difficulty to its brief transitional nature, low support, and overlap with adjacent stages. [1]

10. Separate Standalone / Exhibition Result

The repository also reports a distinct standalone experiment associated with the canonical notebooks 01-05 and the deployable checkpoint. It reports 90.57% accuracy, Cohen kappa 0.8080, macro F1 0.7490, weighted F1 0.9115, best validation kappa 0.8274 at epoch 18/20, and measured CPU latency of 6.2 ms/batch. This experiment should not be merged numerically with the primary person-level cross-validation result because its split and reporting protocol differ. [1]

Metric	Standalone reported value	Why it should be read separately
Accuracy	90.57%	Exhibition/standalone split
Cohen kappa	0.8080	Chance-corrected agreement for that run
Macro F1	0.7490	Equal class weighting; useful under imbalance
Weighted F1	0.9115	Dominated by common classes such as Wake
Best validation kappa	0.8274 @ epoch 18/20	Validation checkpoint selection
CPU latency	6.2 ms/batch	Hardware- and batch-dependent runtime measurement

11. Artifact Audit and Reproducibility Notes

A research paper should make disagreements between public artifacts visible. The current repository README is internally richer than the public landing-page summary, while the hosted Hugging Face model card reports a different evaluation snapshot for the same 99,477-parameter family. The hosted card currently reports 87.7% accuracy, kappa 0.763, macro F1 0.730, weighted F1 89.0%, and per-class F1 values of 0.964/0.445/0.768/0.681/0.771. It also documents the 0.5-35 Hz, 50 Hz notch, z-score preprocessing chain. [2]

These values are not combined with the repository benchmark because they may correspond to a different checkpoint, training artifact, or evaluation snapshot. For publication, the recommended fix is to pin a Git commit and checkpoint hash, regenerate all metrics from one canonical artifact, and export one machine-readable results file that contains dataset manifest, split manifest, model checksum, software versions, and all metrics. The repository already contains protocol-fingerprinting utilities that can support this process. [1]

12. Comparison with Prior Work and Current Research Direction

NeuroSleep sits within a progression of sleep-staging research from convolutional/recurrent models toward highly reusable multimodal and edge-oriented models. DeepSleepNet introduced raw single-channel EEG modeling with CNN and LSTM components; TinySleepNet reduced architectural overhead and emphasized efficient end-to-end training; U-Sleep demonstrated that fully convolutional time-series segmentation can scale to diverse multi-cohort PSG; SleepFM shifted toward large-scale multimodal

representation learning; recent DistillSleep and DistillSleepNet work specifically targets on-device inference and heterogeneous knowledge transfer. [6-11]

System / study	Core design	Scale / setting	Relation to NeuroSleep
DeepSleepNet (2017)	CNN + LSTM on raw EEG	Single-channel EEG	Motivates local feature extraction + temporal context
TinySleepNet (2020)	Efficient CNN + recurrent sequence model	Single-channel EEG; multi-dataset evaluation	Supports the compact-model design philosophy
U-Sleep (2021)	U-Net-like fully convolutional model	15,660 participants across 16 studies	Demonstrates robustness via large heterogeneous training data
SleepFM (2024)	Multimodal foundation representation learning	>14,000 participants; >100,000 hours	Represents the large-scale foundation-model direction
DistillSleepNet (2025)	Heterogeneous multi-level KD + teacher assistant	Sleep-EDF + ISRUC	Shows the current relevance of distillation for deployment
DistillSleep (2025)	Transformer teacher + 109K student	>10,000 recordings; edge devices	Closest deployment-oriented literature context
NeuroSleep (current)	CNN + learnable Gabor + 2-layer GRU	92 eligible records; 52 persons; 4-channel PSG	Compact, multi-modal, causal 300 s context

The comparison is intentionally descriptive rather than a leaderboard: the datasets, channels, label protocols, cohort sizes, training regimes, and metrics differ substantially. A numerical ranking across these systems would not be scientifically defensible without matched-data evaluation. The current literature instead suggests two complementary research directions: scaling models and datasets for generalization, or compressing models for edge inference. NeuroSleep is explicitly positioned in the second direction while retaining multi-modal PSG context. [4,6-11]

13. Model Efficiency and Deployment Analysis

At 99,477 parameters, the model occupies roughly 398 kB of raw FP32 parameter storage before framework overhead. The repository describes it as approximately 400 KB serialized, under 10 MB memory in its deployment configuration, and capable of sub-10 ms CPU inference; the standalone experiment reports a measured 6.2 ms/batch latency. These figures are hardware- and batch-dependent and should be interpreted as engineering measurements, not universal benchmarks. [1]

Resource property	Reported value	Engineering implication
Parameter count	99,477	Small enough for compact inference packaging
Serialized FP32 size	~400 KB	Low model-storage requirement
Input context	300 s	Longer temporal view without a long transformer
Reported CPU latency	6.2 ms/batch in standalone run	Supports interactive/offline CPU inference; exact hardware matters
Deployment memory	<10 MB reported	Potential fit for constrained platforms, subject to runtime footprint
Hosted format	Safetensors checkpoint	Portable and integrity-friendly weight serialization

The main architectural trade-off is visible in the parameter budget: roughly 90.3% of parameters are in the GRU. This means further compression work should first investigate recurrent-state efficiency, sequence length, hidden width, low-rank recurrent approximations, quantization, or replacement of GRU memory with cheaper temporal operators. Compressing the already-small convolutional stem would yield relatively little total savings.

14. Error Analysis and Scientific Interpretation

N1 is the principal failure mode. The stage is both rare and physiologically transitional. Errors are expected to occur around Wake-N1 and N1-N2 boundaries, where the visual differences can be subtle even for human scorers. The reported F1 of 0.445 makes this the clearest target for future modeling work. [1,4]

Wake is highly separable in this cohort. The 0.960 F1 indicates that the model has strong signal for the awake state in the selected data. This should not be interpreted as a guarantee on pathological, noisy, or ambulatory recordings.

REM benefits from multi-modal input. The presence of EOG and EMG provides complementary ocular and muscle-tone cues. The 0.782 F1 suggests that the multi-channel design captures information that a purely single-channel EEG system would not observe directly.

Cross-subject variability is substantial. The confidence intervals for N2-N3-REM are noticeably wider than Wake, indicating that performance varies across held-out people. This reinforces the importance of person-level validation and argues against reporting a single aggregate score without uncertainty.

15. Limitations and Threats to Validity

Limited unique-person count. Although the database contains more recordings, the primary benchmark contains 52 unique persons. Person-level separation is correct, but the sample remains small for broad population claims.

Single-dataset evaluation. No external cohort is reported. Sleep-EDF characteristics, scoring conventions, recording hardware, and demographics constrain conclusions about deployment elsewhere.

Strong class imbalance. Wake dominates the class distribution, while N1 is only ~4.6%. Accuracy can therefore remain high even if minority-stage behavior degrades.

Artifact discrepancies. README, config, and hosted model-card values differ in selected fields. Publication requires a single pinned canonical artifact.

No clinical validation. The system scores sleep stages; it is not validated here for diagnosis or clinical decision-making.

Neuromorphic terminology. Despite the project name, the current network is a conventional artificial neural network. It does not implement an event-driven spiking neural network or neuromorphic hardware execution path. The term should therefore be treated as project branding / design framing unless future versions add genuine SNN components.

No matched ablations. The public artifacts describe the architecture, but a rigorous publication should report controlled ablations for the multi-resolution stem, Gabor branch, GRU depth, and 300-second context.

16. Recommended Publication-Grade Ablation Study

A strong follow-up experiment can make the architecture scientifically defensible without changing the data split. The key requirement is that every variant uses identical subject folds, seeds, preprocessing, optimizer schedule, and stopping rule.

Ablation	Variant	Primary question
A1	Remove Gabor branch	Does explicit learnable spectral structure improve stage discrimination?
A2	Single 25-s stem	How much do fast temporal features contribute?
A3	Single 200-s stem	How much do slow oscillatory features contribute?
A4	Standard convolution instead of depthwise-separable	What is the accuracy/compute trade-off?
A5	1-layer vs 2-layer GRU	Is recurrent depth necessary?
A6	Hidden size 32 / 64 / 96	How does capacity scale with performance?
A7	Context 30 / 90 / 150 / 300 s	What is the minimum useful temporal context?
A8	Unweighted vs weighted loss	How much do N1 and REM weights change minority-stage F1?
A9	Post-training INT8 quantization	Can edge memory/latency improve without unacceptable F1 loss?
A10	External cohort	Does performance transfer beyond Sleep-EDF?

17. Future Work

The next stage of the project should separate three research objectives that are often conflated: accuracy, generalization, and deployment. Accuracy work should focus on minority-stage representations, calibrated uncertainty, and explicit transition modeling. Generalization work should add external datasets and domain-shift analysis. Deployment work should profile CPU, ARM, TFLite/ONNX runtimes, quantization, and memory under realistic hardware constraints.

A particularly promising extension is a teacher-student pipeline in which a high-capacity model learns richer multi-epoch context and the current 99K network acts as the deployable student. Recent literature shows that knowledge distillation is an active strategy for on-device sleep staging. However, any such experiment should preserve the repository's person-level split and must be verified for leakage before its metrics are used as generalization evidence. [9-11]

A second direction is actual neuromorphic computing. If the project retains the NeuroSleep name, a future version could translate the temporal feature extractor or GRU-like memory into spiking neurons, use event-driven encodings, and evaluate energy per inference on neuromorphic hardware. That would turn the current neuromorphic framing into a literal algorithmic claim rather than a naming convention.

18. Conclusion

NeuroSleep is a compact multi-modal sleep-staging model built around a clear engineering objective: retain meaningful physiological and temporal information while keeping the network below 100K parameters. The current ImprovedStudent architecture achieves this through multi-resolution temporal convolutions, depthwise-separable feature extraction, a learnable Gabor spectral branch, and a two-layer GRU covering five minutes of causal context. The repository reports strong Wake performance and useful N2/N3/REM discrimination, while N1 remains the dominant error mode.

The strongest aspect of the project is not a single accuracy number but the combination of person-level evaluation, explicit reproducibility tooling, a compact model budget, and a deployment path. The main scientific work still required for publication is equally clear: unify artifact versions, report controlled ablations, validate externally, quantify calibration and uncertainty, and document the exact preprocessing and checkpoint provenance for one canonical release. With those additions, the project can be evaluated as a serious edge-oriented sleep-staging research system rather than only an exhibition prototype.

19. Reproducibility Checklist

Check	Current status in repository	Publication action
Pinned model architecture	Yes	Include exact Git commit and config in paper supplement
Person-level folds	Yes	Archive the fold manifest
Multiple seeds	Yes: 42/43/44	Publish fold-level raw metrics
Protocol fingerprint	Yes	Publish the generated fingerprint alongside checkpoint
Held-out external dataset	No	Add at least one independent cohort
Ablations	Not yet sufficient in current public benchmark	Run matched ablations
Exact preprocessing in one canonical source	Partly split between README/model card/source modules	Consolidate in one versioned config
Checkpoint/report consistency	Not fully aligned across public artifacts	Regenerate hosted model card and README from same results file
Clinical validation	No	Do not make diagnostic claims until performed

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Appendix A. Architecture Contract

Tensor / module	Shape or setting	Meaning
Input	[B,10,4,3000]	Batch x epochs x channels x samples
Stem output	[B,10,16,~500]	Multi-resolution temporal representation
Pooled features	[B,10,32,8]	Fixed compact representation
Gabor branch	8 filters + 8 -> 16 projection	Learnable spectral descriptors
Fusion	[B,10,272]	Epoch-level feature vector
GRU	2 layers, hidden=64	Inter-epoch memory
Logits	[B,10,5]	Five class scores per epoch
Softmax	[B,10,5]	Stage probabilities

Appendix B. Module-level Design Rationale for Oral Defense

Why four channels? Each modality contributes different physiology: EEG for cortical dynamics, EOG for ocular activity, and EMG for muscle tone. This reduces reliance on one signal family.

Why two convolutional kernels? 25 samples and 200 samples at 100 Hz correspond to 0.25 s and 2 s. Parallel views let the model detect both fast morphology and slower oscillatory structure.

Why depthwise-separable convolutions? They approximate the representational role of ordinary convolution with fewer parameters and less compute, which is aligned with the edge-deployment target.

Why Gabor filters? Sleep scoring is frequency-sensitive. A learnable Gabor bank introduces a direct spectral inductive bias while staying small.

Why adaptive pooling? It compresses the temporal axis to a fixed eight-position budget before spectral fusion, keeping downstream feature width stable.

Why a GRU instead of a large Transformer? The goal is a constrained recurrent memory over only ten epochs. A GRU can capture transition context at much lower parameter cost than a large self-attention stack.

Why weighted loss? N1 and REM are less frequent; weighting their errors prevents the optimizer from being dominated by Wake and N2 examples.

Why person-level folds? A sleep record contains subject-specific signal statistics. Splitting by person prevents those statistics from leaking into the test set.

Why report kappa? Sleep staging is a categorical agreement task and the class distribution is skewed. Kappa captures chance-corrected agreement that complements accuracy.

Why is N1 hard? It is a short transitional state and can look similar to Wake or N2, while representing only a small fraction of epochs in the cohort.

Appendix C. Data and Artifact Provenance

Repository evidence used for the model description: current public README/indexed repository text, project model-card information, and the repository-reported experiment protocol. External evidence was used for Sleep-EDF dataset facts, AASM scoring context, and literature positioning. All claims about project metrics are explicitly presented as repository-reported results rather than independently re-run metrics.

Important versioning note: the public landing page and hosted model artifact are not treated as guaranteed byte-identical releases. A formal publication release should package the exact Git commit, checkpoint, results JSON/CSV, split manifest, and environment lockfile together and regenerate the paper tables from that same bundle.